

Seven gene sets. Seven clear definitions.

Family 1 captures four primary evidence categories. Family 2 contains three CSW-derived GO term groups, with FC0.5 and FC1 provenance shown separately.

4

primary sets

3

GO-derived sets

2

thresholds

7

stocker CSVs

Two families, seven non-interchangeable sets

Each set now receives one slide; overlap is reported separately within each family.

FAMILY 1 · PRIMARY EVIDENCE

- | | | | |
|----|----|--------------------------|---|
| 01 | 6 | Mechanistic | Consistent/correlated + published homeostatic-consistent effect |
| 02 | 20 | Homeostatic genes | Opposing history versus rebound correlations |
| 03 | 97 | CSW 4+ genes | Consistent in four or more of seven datasets |
| 04 | 4 | HLH genes | Potential upstream regulators |

FAMILY 2 · CSW-DERIVED GO

- | | | | |
|----|-----|----------------------|--------------------|
| 05 | 99 | Ribosomal | FC0.5 99 · FC1 12 |
| 06 | 167 | Mitochondrial | FC0.5 167 · FC1 23 |
| 07 | 5 | Immune | FC0.5 0 · FC1 5 |

FC0.5 and FC1 counts are source-provenance counts. A gene may appear at both thresholds.

Mechanistic • 6 genes

Consistent or correlated sleep/wake genes with published effects compatible with homeostatic feedback.

6

paper-highlighted genes

Inclusion requires transcriptomic evidence plus published sleep/wake function.

- Consistent across sleep/wake conditions or correlated with sleep/wake history
- Published perturbation changes sleep/wake
- Effect direction fits a potential homeostatic loop

unc79

NALCN complex

SIFa

neuropeptide

rumpel

glial transporter

AstA-R2

neuropeptide receptor

Trhn

serotonin synthesis

RpL23

ribosome / translation

These six are not the 20 opposing history-rebound correlation genes.

Homeostatic genes • 20 genes

Genes correlated with pre-sampling sleep/wake history and post-sampling rebound in the opposite direction.

15

History+ / Rebound–

Positive correlation with prior sleep;
negative correlation with rebound.

15 • HISTORY+ / REBOUND–

**AstA-R2 • bumpel • CG33080 • CG41378 • CG7460
CG7601 • Chrac-14 • Cyp28c1 • dpr21 • kumpel
Obp44a • Sk1 • Ssk • Trhn • VGlut2**

5

History– / Rebound+

Negative correlation with prior sleep;
positive correlation with rebound.

5 • HISTORY– / REBOUND+

CG42323 • CG7079 • CG9313 • CG9377 • Gtpbp1

CSW 4+ genes • 97 genes

Consistent sleep/wake genes evident across four or more of the seven datasets.

97

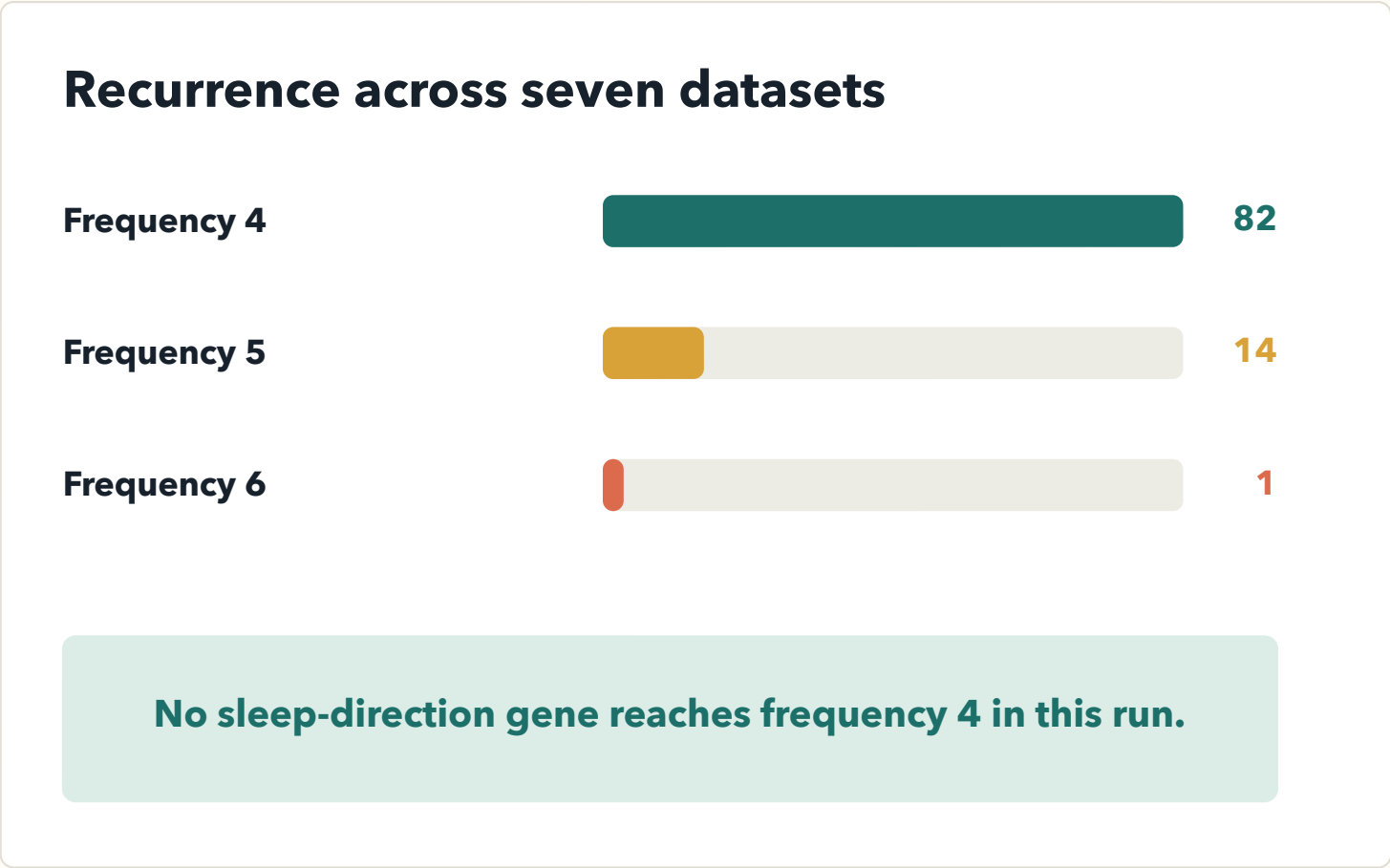
FC0.5 wake genes

The category rule is recurrence across datasets—not GO enrichment.

7

also qualify at FC1

All seven are already contained within the 97-gene FC0.5 set.



HLH genes • 4 genes

Four bHLH factors identified as potential upstream regulators of wake-induced gene expression.

REGULATORY LOGIC

CAGCTG

E-box enriched upstream of
wake-induced genes



**HOMER motif enrichment →
candidate direct regulators**

Results • Table S1

bigmax

FBgn0039509

HLH3B

FBgn0011276

E(spl)m3-HLH

FBgn0002609

E(spl)mbeta-HLH

FBgn0002733

**This set is regulatory evidence—not frequency or correlation
evidence.**

Ribosomal / translation · 99 genes

GO term-grouped hit genes with FC0.5 and FC1 source provenance shown separately.

THRESHOLD
PROVENANCE

99

unique genes in this set

FC0.5

99

genes supported

FC1

12

genes supported

BOTH

12

appear at both thresholds

99 FC0.5 + 12 FC1 – 12 both = 99 unique

Most frequent approved
terms

Ribosome

93

Structural constituent of ribosome

92

Translation

77

Cytoplasmic translation

65

Cytosolic ribosome

58

87 FC0.5-only · 0 FC1-only · 12 at both thresholds

Mitochondrial / metabolism · 167 genes

GO term-grouped hit genes with FC0.5 and FC1 source provenance shown separately.

THRESHOLD
PROVENANCE

167

unique genes in this set

FC0.5

167

genes supported

FC1

23

genes supported

BOTH

23

appear at both thresholds

167 FC0.5 + 23 FC1 – 23 both = 167 unique

Most frequent approved
terms

Mitochondrion

121

Mitochondrial inner membrane

72

Oxidative phosphorylation

46

Mitochondrial translation

30

Mitochondrial matrix

24

144 FC0.5-only · 0 FC1-only · 23 at both thresholds

Immune • 5 genes

The immune GO term group is FC1-exclusive in this analysis.

THRESHOLD
PROVENANCE

5

unique genes in this set

FC0.5

0

genes supported

FC1

5

genes supported

BOTH

0

appear at both thresholds

0 FC0.5 + 5 FC1 – 0 both = 5 unique

Five FC1-
supported genes

BomBc2

BomS1

BomS2

IM14

Thor

Approved GO terms

Defense response

4

Antibacterial humoral
response

4

Response to bacterium

4

0 FC0.5-only • 5 FC1-only • 0 at both

Only three genes cross primary-set boundaries

Overlap is calculated among Mechanistic, Homeostatic, CSW 4+, and HLH only.

Observed shared genes

	Mechanistic × Homeostatic	2	Trhn • AstA-R2
	Mechanistic × CSW 4+	1	RpL23
	All other pairings	0	No shared FBgns

Family 1 union

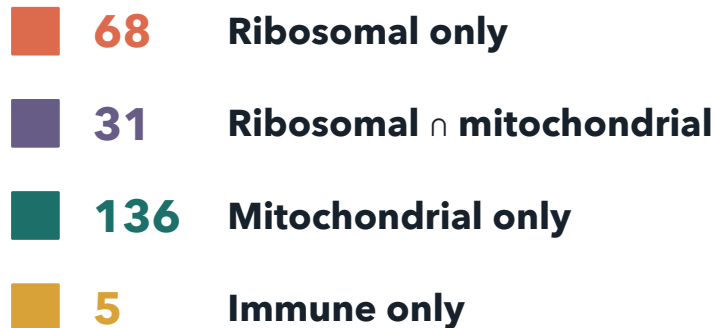
124	unique FBgns
121	in exactly one set
3	in exactly two sets
0	in three or four sets

HLH genes are fully disjoint.

The GO-derived union contains 240 unique genes

99 ribosomal + 167 mitochondrial + 5 immune – 31 shared ribosomal/mitochondrial genes = 240 unique FBgns.

GO term-group membership • 240 unique genes



Category overlap

31 genes retained in both ribosomal and mitochondrial buckets

Threshold provenance across the 240-gene union

203 FC0.5 only

32 both thresholds

5 FC1 only

Only the five immune genes are FC1-exclusive.